

KABIR KAKKAR

+1-412-606-1140 | kabir.kakkar.cmu@gmail.com | </in/kabir-kakkar/> | </kabir-kakkar>

Summary

Data Engineer with hands-on experience designing production-scale batch and streaming pipelines across Kafka, Spark, Flink, and Airflow, and building cloud data warehouses on Snowflake, Redshift, and BigQuery. Skilled at architecting reliable, cost-efficient ETL/ELT systems that process hundreds of gigabytes to hundreds of millions of events daily, with a track record of measurably improving latency, data quality, and pipeline reliability.

EDUCATION

Carnegie Mellon University

Master of Information Systems Management

August 2024 - December 2025

Manipal University Jaipur

Bachelor of Technology, Computer and Communication Engineering

June 2017 - June 2021

SKILLS

Data Engineering & Streaming: Kafka, Spark (PySpark), Spark Streaming, Flink, Airflow, Hadoop, dbt, ETL, Databricks, Kinesis

Databases & Warehousing: Snowflake, Redshift, BigQuery, PostgreSQL, MongoDB, Redis, Pinecone (Vector DB), Data Lake, Delta Lake, Apache Iceberg, Apache Beam, Data Modeling

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, CI/CD, Prometheus, Grafana, AWS Glue, AWS EMR, GitHub Actions

Languages: Python, SQL, Java, JavaScript, TypeScript, C++, Linux / Bash

Backend & APIs: FastAPI, Flask, Django, Spring Boot, REST APIs, GraphQL, Microservices, Git

PROFESSIONAL EXPERIENCE

MedSarathi LLC

Software Development Engineer - Data Engineer , Internship

July 2025 - August 2025

- Designed and built a data ingestion pipeline that parsed, cleaned, and structured unstructured CV/document data into PostgreSQL, standardizing inconsistent source formats into a queryable schema with 95%+ extraction accuracy
- Built the retrieval infrastructure for a hybrid RAG system - embedding generation, vector indexing in Pinecone, and semantic search across 100+ program records - enabling end-to-end recommendation delivery in under 3 seconds
- Engineered backend data infrastructure for document ingestion, storage, and observability, supporting over 10,000 daily processing requests with Dockerized deployment for reliable, repeatable pipeline execution
- Integrated structured applicant data with the retrieval layer to feed a downstream ranking and recommendation engine, ensuring low-latency, high-integrity data flow across the pipeline

Accenture

Software Engineer - Data Engineer

May 2021 - May 2023

- Designed and maintained scalable ETL pipelines using Apache Spark and Python on AWS Glue, processing 500GB+ of data daily and improving data availability by 30% for downstream analytics teams
- Built and optimized data warehouse solutions on Amazon Redshift, implementing partitioning and query tuning strategies that reduced average query latency by 40%
- Migrated legacy on-premise data infrastructure to the cloud, improving system scalability and reducing operational costs by 20%
- Automated data workflows using Apache Airflow, cutting manual data processing effort by 15+ hours per week and improving pipeline reliability

PROJECTS

Real-Time Anomaly Detection Pipeline for Financial Fraud

- Designed a Kafka + Flink streaming pipeline processing 15K+ events/sec with exactly-once checkpointing and event-time watermarking, reducing fraud detection latency to under 2 seconds
- Built a Redis-backed real-time feature store integrated into a stateful Flink job, cutting feature-lookup latency by 90% and supporting sub-second decisioning at production scale

E-Commerce Clickstream Lakehouse

- Architected a medallion lakehouse (Bronze/Silver/Gold) on Apache Iceberg processing 100M+ daily clickstream events, using PySpark and Kafka for ingestion and automated compaction to cut small-file count by 95%
- Built incremental Merge-based transformations and dbt Gold-layer models with 150+ automated tests, reducing reprocessing cost by 80% and catching data quality regressions before they reached BI dashboards

Cloud Analytics Warehouse (dbt)

- Designed a Kimball-style star schema in BigQuery with SCD Type 2 dimension tracking across 500K+ records, enabling accurate historical trend analysis for business stakeholders
- Built 40+ dbt models with 150+ automated tests (uniqueness, referential integrity, freshness) and a CI pipeline gating pull requests, reducing undetected data quality issues by ~70%